

LESSON 10.5

INTERPRETING GRAPHS OF FUNCTIONS



LESSON INTRODUCTION

MA.8.F.1.3 Analyze a real-world description or graphical representation of a functional relationship between two quantities and identify where the function is increasing, decreasing, or constant.

LEARNING TARGETS

- I can identify where a function is increasing, decreasing, or constant from its graph.
- I can write a description of a graphical representation of a function in context.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can a graph tell you a story of a relationship between two quantities?
- What is the difference between speed and velocity, and how is that related to the slope of a line?

LESSON NARRATIVE

Graphical representations provide a visual story of relationships between quantities. This lesson begins the process of analyzing, describing, and interpreting graphical representations of various relationships. Students will also be introduced to the connection between the slope of a line, speed, and velocity.

COMPONENT SUMMARY

10.5.1 WARM-UP (~5 MIN.)

Which One Doesn't Belong?

10.5.3 BRING IT TOGETHER (5-10 MIN.)

Connect the slope of a line to speed and velocity

10.5.5 WRAP-UP (5-10 MIN.)

Match graphs to descriptions and determine possible

10.5.2 EXPLORATION (20-25 MIN.)

Describe, analyze, and draw graphs that represent a turtle's journey

10.5.4 GUIDED PRACTICE (10-15 MIN.)

Analyze and interpret graphs of functions

independent and dependent variables

RESOURCES**GLOSSARY**

Key Terms:

- Speed
- Velocity

Additional Terms: N/A

MATERIALS

- Graph paper
- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS**COMMON MISCONCEPTIONS**

- Students may confuse the concepts of velocity and speed, verbally mixing up the values or statements.
- Students may misinterpret speed and location on a graph when interpreting slope.
- Students may misinterpret straight lines in the turtle activity as the turtle walking straight (direction).

THINGS TO CONSIDER

- Consider incorporating project-based learning as an extension later in the unit, specifically referencing student interpretations from this lesson.
- Consider talking to the science department on how they approach speed versus velocity.

LITERACY AND LANGUAGE ROUTINES**Which One Doesn't Belong?**

In the 10.5.1 Warm-Up, students are presented with four graphs and asked to determine which graph does not fit in with the other three graphs. The routine highlights the low-floor, high-ceiling approach to interpreting graphs, with students able to answer with informal language while simultaneously offering opportunity to engage in higher-order thinking.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD**MA.K12.MTR.2.1 Multiple Representations**

Students have the opportunity to practice interpreting and representing math in real-world contexts using verbal descriptions, written statements, and graphical representations. Students make sense of speed, velocity, rate, time, distance, and their relationships in a variety of ways to intentionally target the multiple representations in context.

DIFFERENTIATE INSTRUCTION

In the 10.5.1 Warm-Up, encourage students to give mathematical reasoning for their choice of which graph does not belong. Accept and allow students to provide any nonmathematical reasons that can then be supported mathematically.

All components in this lesson provide a visual entry point because the graphs provide visual representations of the relations. Whole-class synthesis discussions will provide an auditory entry point. Having students model the movement of the turtle by walking across the front of the room can support kinesthetic learning.

Many components ask students to write a description of a graphical representation of a function in context. Consider allowing students to share their answers verbally before having them provide a written explanation.

10.5.1 WARM-UP: WHICH ONE DOESN'T BELONG?

QUESTIONING

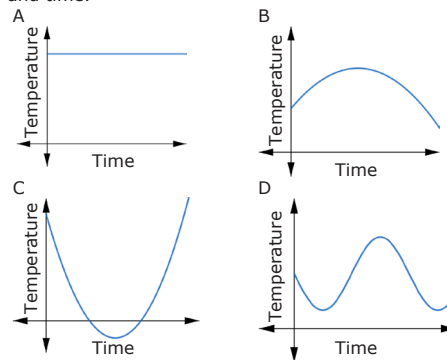
Challenge students to think of at least one reason to justify why each graph does not fit with the other three graphs.

Ask students to describe a story that could match the changes in temperature over time shown in each graph. (Notice how this reflects the 10.5.5 Wrap-Up.)



10.5.1 Warm-Up: Which One Doesn't Belong?

Four graphs show the relationship between temperature and time.



1. Which one doesn't belong?

Answers will vary.

2. Explain your choice.

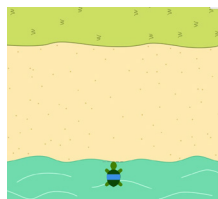
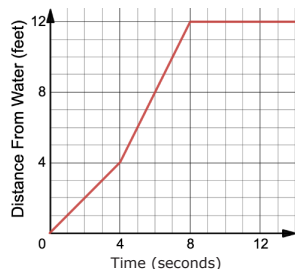
Possible reasons for all graphs:

- A- The temperature remains constant whereas it is changing in the other graphs; It's the only line.*
- B- The temperature increases first for half the time before decreasing the rest of the time.*
- C- The graph includes negative temperatures (temperature is below the x-axis for some times); The temperature decreases first for half the time before decreasing the rest of the time.*
- D- The temperature is increasing and decreasing regularly over time.*



10.5.2 Exploration: Distance Versus Time Graphs

1. A turtle is traveling across the sand.
 - a. Draw a graph using line segments to represent a possible path the turtle could take.



Student graphs will vary. They may include areas of zero slope and should not include any vertical lines.

- b. Write a brief story describing the turtle's journey based on your graph.
Answers will vary. The turtle maintained a slow, steady pace as he traveled further from the water.
 - c. Share your graph and story with your partner. Discuss any differences in your graphs or interpretations. Make revisions to your story or graph, if necessary.

10.5.2 EXPLORATION: DISTANCE VERSUS TIME GRAPHS

ENRICHMENT

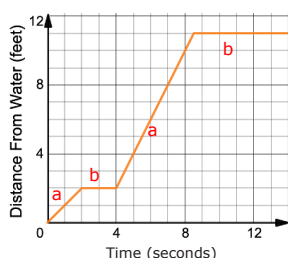
Anticipate where students might struggle making connections between their story (written description) and their graph (visual representation) or need additional support in depicting their story.

- What is happening to the distance from the water when the turtle is walking away? How would this look on a graph?
- What is happening to the distance from the water when the turtle is standing still? How would this look on a graph?
- What is happening to the distance from the water when the turtle is walking toward the water? How would this look on a graph?

TEACHER MOVES

While students are working, make note of interesting stories about the graph. After question 1b, consider having those individuals share their stories with the whole class, as some students benefit from hearing multiple interpretations.

2. Luca drew the graph shown to represent a new turtle's journey across the sand.



- a. On your own, label the graph as described below.
 - Label any parts of the graph where Luca's turtle is walking away from the water with the letter *a*.
 - Label any parts of the graph where Luca's turtle is standing still with the letter *b*.
 - Label any parts of the graph where Luca's turtle is walking toward the water with a letter *c*.
Shown on graph.
 - b. Compare your labels with your partner and resolve any differences, if necessary.
3. Work with your partner to answer the following using Luca's graph.
 - a. What is the turtle's distance from the water at 8 seconds?
The turtle's distance from the water is 10 feet.
 - b. When is the turtle's distance from the water 4 feet?
The turtle's distance from the water is 4 feet at 5 seconds.
 - c. Based on the graph, the turtle is walking between 0 and 2 seconds, and between 4 and 8.5 seconds. Determine whether the turtle is traveling faster between 0 and 2 seconds, or between 4 and 8.5 seconds.
The turtle is traveling faster between 4 and 8.5 seconds. Between seconds 4 and 8.5 is a rate of change of 2 feet per second and between seconds 0 and 2 is a rate of change of 1 foot per second.

DIFFERENTIATE INSTRUCTION

Perhaps the most common misconception is that lines and points represent location and distance. For example, students may believe a flat or straight line indicates the turtle is walking straight.

Students can walk to model the graph. Some students may think that their movements will be the same as the graph. Tell the volunteers that they have to walk in a straight line. Then ask students if the distance the turtle walked had to be in a straight line.

Use scaffolded prompts to position students in applying MA.K12.MTR.2.1: Multiple Representations to make sense of information.

What does a horizontal line say about the turtle's distance? Does time pass on a horizontal line? (A horizontal line indicates that the turtle is not moving. Time does pass; for example, the turtle is at the same distance for 10 and 11 seconds.)

Take Luca's example. Based on your response, tell the story of what happened for the first four seconds.

ENRICHMENT

On a graph, how can the steepness of a line, or slope, be used to make comparisons?

10.5.2 EXPLORATION: DISTANCE VERSUS TIME GRAPHS (CONTINUED)

DIFFERENTIATE INSTRUCTION

Think-Pair-Share

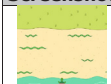
Notice the intentional scaffolding and multiple entry points. Give students the opportunity to first visually process the images. Prompt students to make conjectures or begin mentally sketching what the graph might look like. Students then take those conjectures and represent those images on a coordinate plane, making sense of axes and rates. Students finish by refining their responses with a partner in question 4b.

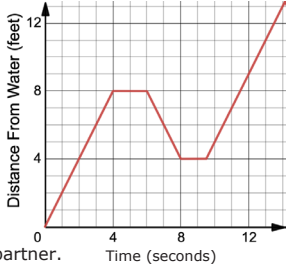
ENRICHMENT

While students are discussing in question 4b, monitor for common misconceptions. Consider incorporating prompting questions that connect this activity to the formal vocabulary in the next activity.

- What is happening to the distance as the turtle walks forward, away from the water?
- What is happening to the distance when the turtle stops walking?
- What is happening to the distance when the turtle backs up? Does the turtle have to back up, or could the turtle have turned around?
- When using the word “hurries,” what do you think this means regarding the turtle’s speed? How do you think a change in speed will look on a graph?

4. Four screenshots and brief descriptions from an animation are shown. In the animation, a turtle is being careful to avoid crawling snakes as it walks across the sand.

Screenshot	Description
	The turtle begins in the water.
	Avoiding the crawling snakes, the turtle walks forward at a constant rate until the point shown, where he stops for a few seconds while a snake passes by.
	Then, the turtle backs up to avoid another snake. He waits a few seconds there.
	The turtle then hurries across the rest of the sand to safety.

- a. On your own, draw a graph of the turtle’s distance from the water over time based on the description in the table. *Answers will vary.*
- 
- b. Compare your graph with your partner. Discuss your graphs and make any changes, if necessary.



10.5.3 Bring it Together: Distance Versus Time Graphs

In graphs comparing time as the independent variable and distance as the dependent variable, the slope of the line represents **velocity**.



Speed vs. Velocity

Speed is the rate at which an object is moving in units of distance per unit of time.
Velocity is the rate *and* direction of an object's movement.

The slope of a line gives both the rate of change and the direction of the change, positive or negative, so it can be used to determine both the speed and the velocity of an object.

1. Complete each statement.

a. On a distance-time graph, a horizontal line

means ☐ the object is not moving.
☐ the object is moving at a constant speed.

b. On a distance-time graph, a diagonal line

means ☐ the object is either speeding up or slowing down.
☐ the object is traveling at a constant speed.

2. How can the speed of the object be determined for a line segment on a distance-time graph?

The speed of an object is found by finding the rate of change, or slope. The rate of change is represented by change in distance divided by the change in time.

10.5.3 BRING IT TOGETHER: DISTANCE VERSUS TIME GRAPHS

ENRICHMENT

Have students assimilate this new information into their existing schema through the lens of the turtle activity. What did this look like for the graph of the turtle? Use words like “speed” or “velocity” in your response.

10.5.4 GUIDED PRACTICE: INTERPRETING GRAPHS OF FUNCTIONS

DIFFERENTIATE INSTRUCTION

If necessary, provide this language on the board through the lens of the turtle activity as scaffolding. “The turtle’s distance from the water is a function of time.”

ENRICHMENT

Ask students while circulating to identify the dependent and independent variables as a quick pulse check in preparation for 10.5.5 Wrap-Up.

TEACHER MOVES

Students sometimes confuse a graph that represents distance over time as the path that an object will take. Consider targeting this misconception by spiraling the questions from the 10.5.2 Exploration 2b and having students apply those responses here.

- Does the arc of this graph represent the path of the bottle rocket? Why or why not?

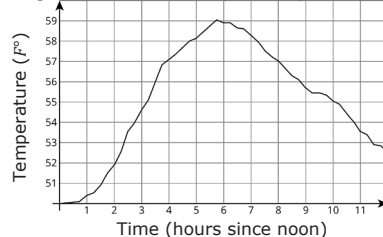
Think about the turtle.

- What did the straight line mean? That the turtle was walking straight? Why not? (Represented passing time and distance.)
- What does that mean in this context? Is this the path of the rocket or something else? (It represents the relationship between the time that has passed and the height from the ground.)

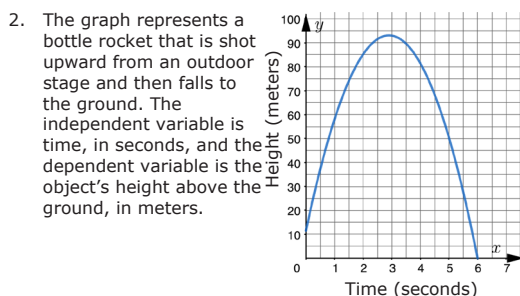


10.5.4 Guided Practice: Interpreting Graphs of Functions

- The graph shows the temperature between noon and midnight one December day in Freeport, Florida.



- Complete the statement.
In this relationship, ☐ time ☒ temperature is a function of ☐ time. ☒ temperature.
- Was it warmer at 3:00 p.m. or 9:00 p.m.? 9:00 pm
- Approximate the time the temperature was at its highest.
The temperature was at its highest at approximately 5:45 PM.
- Complete the statement.
When the input for the function is 8, the output is 57. This means that at 8:00 P.M., the temperature was 57°F in Freeport, FL.



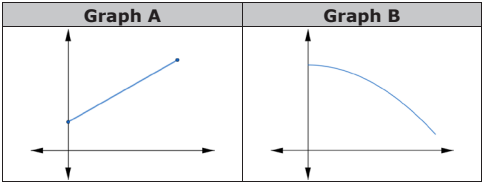
- Complete the description of the story told by the graph.
The bottle rocket is launched from an outdoor stage that is 10 meters above the ground. It continues flying upward for about 2.8 seconds until it reaches approximately 93 meters above the ground. The rocket then begins to fall, landing on the ground after about 6.1 seconds.
- Complete the statement and the inequality.
The domain of the function that represents the rocket’s path is between 0 and 6.1 seconds.

$$0 \leq x \leq 6.1$$



10.5.5 Wrap-Up: Ticket Out the Door

1. Two graphs are shown.



a. Complete the table by writing the letter of the graph that matches each description. Then, name possible independent and dependent variables that could be used to label the axes.

Description	Graph	Independent Variable	Dependent Variable
A plant grows the same amount every week.	A	Time (Weeks)	Plant Height
The day started very warm but then it got colder.	B	Time (Hours)	Temperature

b. Complete the statement.

Graph

☐ A

☐ B

 shows the function

☐ decreasing

☐ increasing

 over the interval shown to model how the

☐ number of weeks

☐ height of the plant

 changes over the

☐ number of weeks.

☐ height of the plant.

10.5.5 WRAP-UP: TICKET OUT THE DOOR

QUESTIONING

Consider a challenge question that extends student thinking to predict how changing information can change the graph or model. Prompt students to create a situation in which the opposite is true for each graph. For example, although Graph A could represent the plant growth each week, if the axes were inverted, the graph might look like the growth is decreasing each week even though it is increasing over time.